

M.S. THESIS PROPOSAL

A Bayesian Framework to Model Transmissible Cancer Dynamics

within *Mya arenaria* Populations

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Abstract

Bivalve Transmissible Neoplasia (BTN) is an increasingly prevalent cancer spreading among bivalve species worldwide. BTN introduces complexities not common in many other infectious diseases due to its marine environment. Therefore, little is known about its transmission dynamics and population effects. This study develops a Bayesian framework to model BTN spread within *Mya arenaria* populations to address key gaps in our understanding, with a focus on statistical methodology for parameter inference and model development. We use a Bayesian compartmental modeling approach to infer and refine model parameters and leverage controlled laboratory and survey data. Laboratory data provides information on cancer cell emission, disease progression, and environmental effects. Survey data from the field includes samples from East Coast sites that are used for fitting the model to disease progression over time in its natural environment. To encompass the intricacies of BTN, our model framework builds on the traditional Susceptible-Exposed-Infectious (SEI) epidemiological model by including cancer particle survivability components and the environmental influence of water temperature.

Keywords: Bivalve Transmissible Neoplasia, Bayesian framework, compartmental modeling

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Introduction

My research will address knowledge gaps by developing a marine disease model that captures the unique complexities of BTN and provides an accurate prediction of the disease spread.

Research Goal. To develop a quantitative marine disease model specific to BTN in *Mya arenaria* (soft-shell clam) populations to study its spread and dynamics over time through a Bayesian compartmental modeling approach, leveraging controlled lab data and field survey data.

Objectives. The specific objectives of this study are to (1) quantify the fundamental transmission parameters of BTN in East Coast *M. arenaria* populations, (2) incorporate cancer particle dynamics into the transmission model to account for BTN's unique transmission vector, and (3) determine the impact of temperature on disease progression, exploring seasonal variation in the disease spread.

Significance. By developing a comprehensive marine disease model for BTN, this research will provide new insights into the spread and dynamics of transmissible cancers in marine organisms. The model will enhance our understanding of how environmental factors, such as temperature and the survivability of cancer cells in the water column, influence disease transmission. This has broader implications for the management of bivalve populations and other marine diseases, as well as advancing knowledge on transmissible cancers.

Literature Review

This literature review provides an overview of research related to BTN, marine disease modeling, and Bayesian approaches in epidemiological modeling. Understanding this literature will inform the development of a transmission model for BTN's spread within *M. arenaria* populations and quantify its ecological impacts.

Bivalve Transmissible Neoplasia. Transmissible cancer is a rare phenomenon that naturally occurs in a few species, including Tasmanian devils, dogs, and several bivalve species (Metzger & Goff, 2016). Devil Facial Tumor Disease (DFTD) (Pearse & Swift, 2006) and Canine Transmissible Venereal Tumor (CTVT) were identified as transmissible in 2006 (Murgia et al., 2006). More recently, in 2015, a fatal leukemia-like cancer affecting bivalve species worldwide, including soft-shell clams, was confirmed to be transmissible (Metzger et al., 2015). Due to its recent classification, much remains unknown about BTN, particularly its mechanisms of spread among individual bivalves and populations.

BTN is spread via transmissible cells derived from a single clam (Metzger et al., 2015). The cancer cells are released by infected clams and can survive for weeks in seawater, with colder temperatures being more suitable for their survivability (Giersch et al., 2022). Studies also show a seasonal pattern of BTN prevalence, with higher incidence rates occurring during late fall and winter compared to early summer each year (Brousseau, 1987; Leavitt et al., 1990).

Marine Disease Modeling. While infectious diseases occur in all ecosystems, most epidemiological models are derived from terrestrial environments. Marine diseases, however, exhibit different dynamics, such as variations in parasite transmission and recruitment patterns (McCallum et al., 2004). Adapting current epidemiological models to marine environments is critical for generating accurate disease predictions, especially as marine ecosystems continue to change rapidly (Lafferty, 2017).

Various environmental stressors, including climate change, influence the ocean's conditions, which in turn affect disease outbreaks (Tracy et al., 2019). Temperature, in particular, can accelerate disease spread and can also explain how pathogens persist in a marine system, but it can also contribute to the decline of pathogens (Aalto et al., 2020). Data-driven modeling and

seasonal forecasts are essential components of disease response strategies, enabling early intervention and effective disease surveillance (Groner et al., 2016).

Bayesian Approaches in Disease Modeling. Bayesian analysis offers a powerful method for making inferences and forecasting disease spread, especially with limited data (Cook et al., 2023; Grinsztajn et al., 2021). By combining prior knowledge with available data, Bayesian frameworks allow for probabilistic inference, yielding posterior distributions that quantify uncertainty in model parameters and predictions (Gelman et al., 2020). The computational tools provided by Stan, such as Hamiltonian Monte Carlo (HMC), enable efficient exploration of complex prior distributions (Carpenter et al., 2017). This approach is particularly useful for estimating key parameters governing disease progression, such as the basic reproduction number (R_0), which helps assess the potential for BTN spread within populations (Kermack et al., 1997).

Methodology

The end goal for the model (Figure 1) is formulated for sessile invertebrates with the filtration of waterborne pathogens and is built up from simpler SEI models (Bidegain et al., 2016).

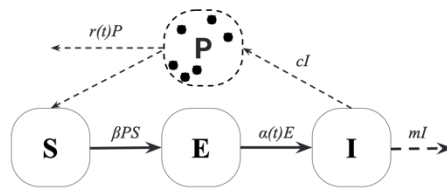


Figure 1: Marine infectious disease model adapted from Bidegain et al. 2016, where β is the disease transmission rate, α is the latent period, m is the disease mortality, c is the release rate of infective particles by infectious animals, r is the inactivation rate of particles in the water column and t represents temperature dependent parameters.

Proposed Chapters.

- I. *Model Development and Parameter Estimation.* In this section, I will present the framework for my epidemiological model. I will describe the disease model, equations, and underlying assumptions and will use laboratory data to inform prior distributions for model parameters.
- II. *Fitting Model to Survey Data for Posterior Estimation.* In this section, I will validate the model using survey data, providing model diagnostics and posterior estimates. This will include deriving the R_0 value and making predictions about disease spread based on the model.

Data. To model the transmission dynamics of BTN in *M. arenaria* populations, I leverage both controlled laboratory experiments and field survey data. Laboratory data from the Metzger Lab provides detailed insights into cancer particle emission, disease progression (Figure 4), and the [environmental effects](#) on cancer dynamics. Field survey data, collected bi-monthly from East Coast clam populations, offers realistic observational data for fitting the model. This data includes approximately 50 samples per site from multiple locations in Massachusetts and Maine, collected by Gloucester Marine Genomics Institute (GMGI) and Bigelow Laboratory for Ocean Sciences (BLOS), as shown in Figure 2. Additionally, eDNA samples from the seawater at the sites are collected, allowing for the quantification of cancer cells in the water over time.

Note: The Metzger Lab has also conducted surveys throughout the Puget Sound. Specifically, there is similar data for BTN spread among Cockles in Penn Cove. At this site, there are 30 samples taken every other month for two years. Applying my model formulation to this data of another species would be an interesting application of my model and its performance in predicting the spread of BTN.

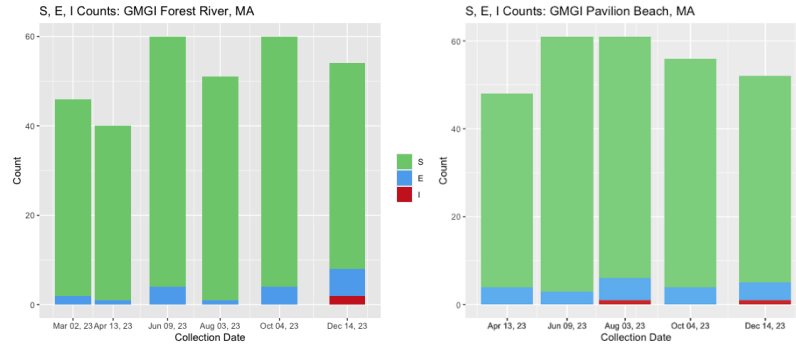


Figure 2: Bi-monthly count survey data from 2023 of East Coast clams.

Progress. A key milestone has been the development of the basic SEI model (Figure 3) (Kermack et al., 1997) for *M. arenaria* in Forest River, MA, based on 2023 survey data, as well as progression data from the lab. This was objective (1) outlined in the introduction.

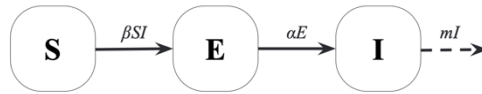


Figure 3: Basic SEI model, where β is the disease transmission rate, α is the latent period, and m is the disease mortality rate.

Time of Infectious. First, I calculated the point at which clams become infectious based on eDNA and hemolymph samples from individual clams. At 24% cancer in the hemolymph, there was a significant point of change in the emission of cancer particles into the water, as determined by eDNA tank samples.

Progression Parameter Inference. Using this information, I was able to differentiate between exposed and infectious clams. Next, I calculated the progression parameters for the basic disease model. Using longitudinal data (Figure 4), I calculated parameter estimates for α and m . For α , I included samples that started with less than 10% cancer and reached 24%. For m , I only included samples that began with less than 24% cancer and died above the threshold.

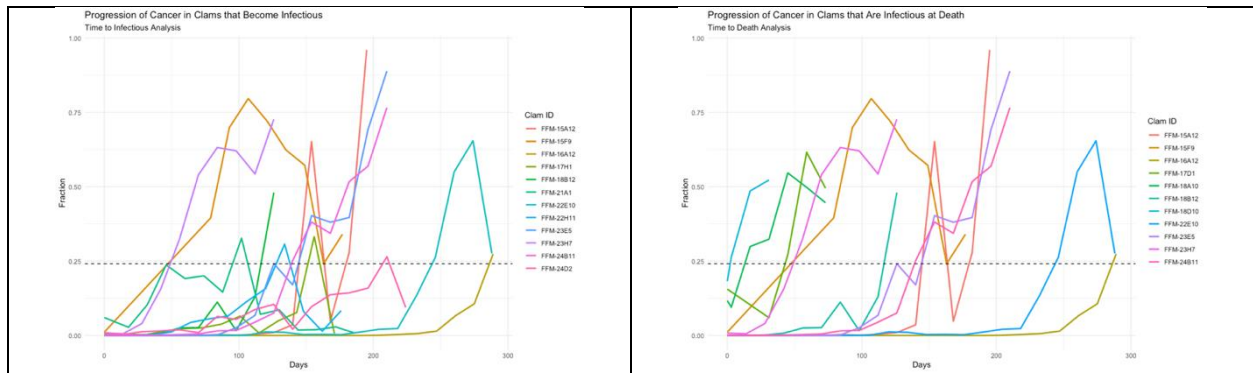


Figure 4: Data for progression from exposed to infectious (left) and from infectious to death (right) with the infectious threshold of 0.24 represented by the dotted line.

I formulated a Bayesian hierarchical model to fit a modified Gompertz curve (Zwietering et al., 1990) to each sample. This is a commonly used model for tumor growth. I assumed that cancer progresses similarly across all clams, with group-level parameters for the shape of the curve and individual-level parameters for the time of progression.

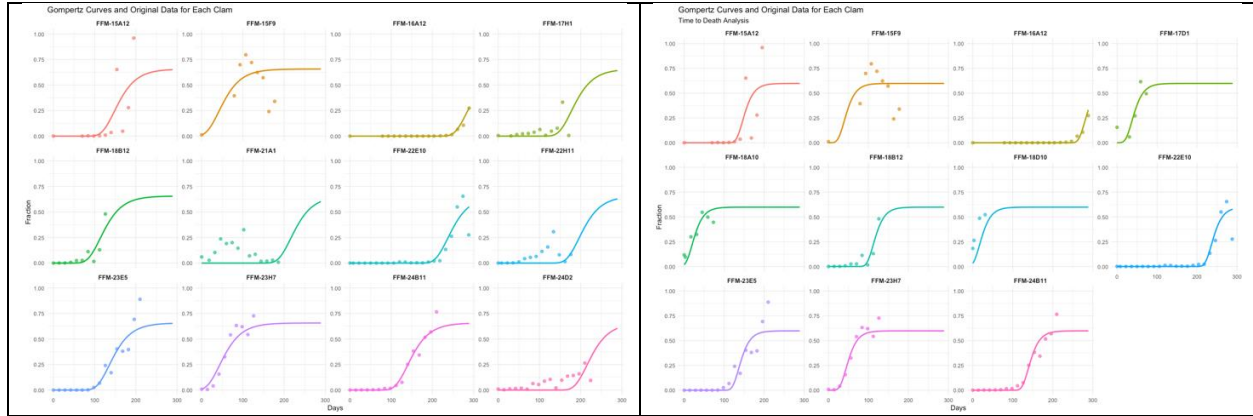


Figure 5: Gompertz fits for each clam sample for α analysis (left) and m analysis (right).

Using these fits, I calculated the prior distributions for the basic SEI model. Since the fitted curves do not reach zero, and the point of exposure is unknown, I defined a lower limit for the time to the infectious state by subtracting the time of the lowest detectable amount of cancer from the time of being infectious for each clam. The average value across all iterations was 80.6 days, and this lower limit was incorporated into the prior distribution for α . For m , I used a gamma distribution to model the time to death (Figure 6), and across all iterations, m had an average of 58.58 days.

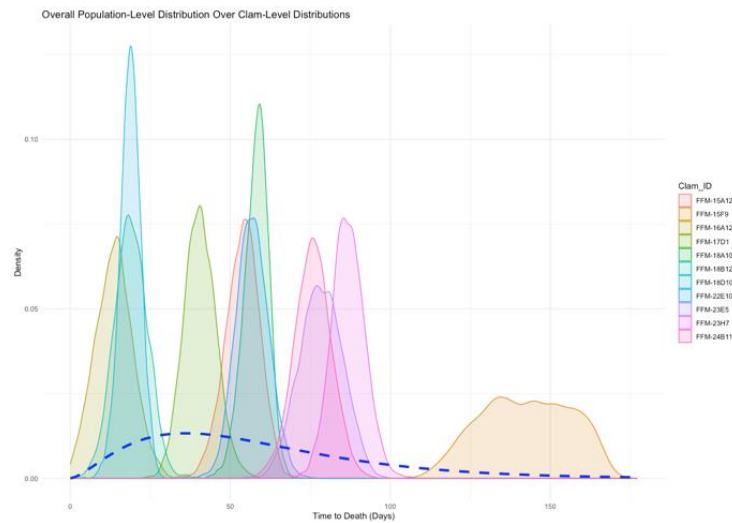


Figure 6: Time to death estimate.

I incorporated these priors into the SEI model and used the 2023 GMGI Forest River survey data to obtain the final parameter estimates (Figure 7).

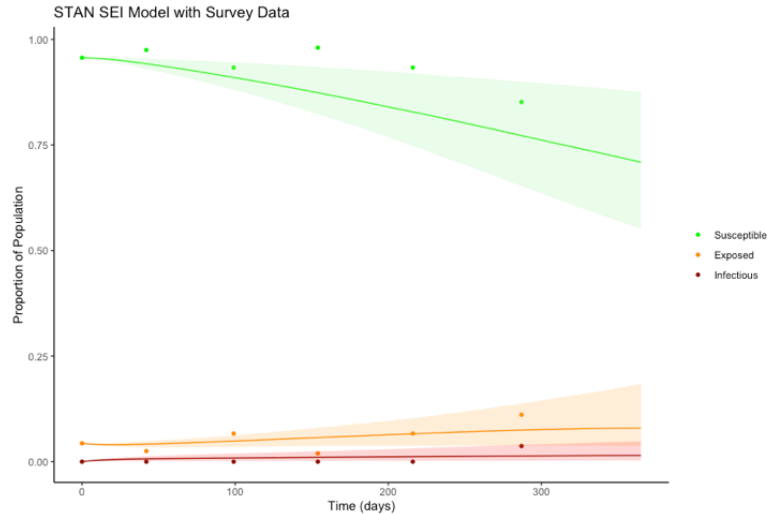


Figure 7: Basic SEI model run in STAN with 2023 GMGI Forest River survey data. The lines represent the model's fit with the estimated parameters, while the dots correspond to survey data points.

Next Steps. I need to make some final adjustments to the model shown in Figure 7 to ensure a strong foundation for further development. This includes enabling the model to estimate the initial values for disease progression and assessing additional diagnostics to validate the model's accuracy. I also plan to run the model with one more year of survey data to refine the parameter estimates. With the inclusion of more data, I expect to observe a cyclic pattern in the disease dynamics, where temperature data could help explain these fluctuations. With the finalization of the basic SEI model, I can expand it to incorporate complexities such as particle transmission dynamics (objective 2) and temperature effects (objective 3), which will guide the next steps in model development.

Adding the cancer particle compartment and the associated differential equations to the model in STAN should be a straightforward integration. The next step will be running the model with the cancer particle eDNA data from the survey sites, which serves as a proxy for the amount of cancer particles in the water throughout time. I will conduct similar Bayesian analyses to inform the model of the transmission rate of cancer particles into the water column from an infectious clam, using the available lab data. Once the particle component is successfully integrated, I will proceed with incorporating the effects of temperature. Temperature logs are available for each site over the survey period. There is also lab data for both disease progression in clams and cancer particle inactive rates under varying temperatures to help inform the temperature-dependent parameters in the model, α and r .

Timeline

Date	Milestone
March 3, 2025	Prepare proposal
March 12, 2025	Submit Master's Thesis Proposal form with committee approval
November 2025	Submit first draft of the thesis to committee and for publication
February 2026	Submit near-to-final draft of the thesis to committee
February 2026	Submit Request For M.S. Final Examination form
March 2026	Final Examination

References

- Aalto, E. A., Lafferty, K. D., Sokolow, S. H., Grewelle, R. E., Ben-Horin, T., Boch, C. A., Raimondi, P. T., Bograd, S. J., Hazen, E. L., Jacox, M. G., Micheli, F., & De Leo, G. A. (2020). Models with environmental drivers offer a plausible mechanism for the rapid spread of infectious disease outbreaks in marine organisms. *Scientific Reports*, *10*, 5975. <https://doi.org/10.1038/s41598-020-62118-4>
- Bidegain, G., Powell, E. N., Klinck, J. M., Ben-Horin, T., & Hofmann, E. E. (2016). Marine infectious disease dynamics and outbreak thresholds: Contact transmission, pandemic infection, and the potential role of filter feeders. *Ecosphere*, *7*(4), e01286. <https://doi.org/10.1002/ecs2.1286>
- Brousseau, D. J. (1987). Seasonal aspects of sarcomatous neoplasia in *Mya arenaria* (soft-shell clam) from Long Island Sound. *Journal of Invertebrate Pathology*, *50*(3), 269–276. [https://doi.org/10.1016/0022-2011\(87\)90092-9](https://doi.org/10.1016/0022-2011(87)90092-9)
- Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). *Stan*: A Probabilistic Programming Language. *Journal of Statistical Software*, *76*(1). <https://doi.org/10.18637/jss.v076.i01>
- Cook, J. D., Williams, D. M., Walsh, D. P., & Hefley, T. J. (2023). Bayesian forecasting of disease spread with little or no local data. *Scientific Reports*, *13*(1), 8137. <https://doi.org/10.1038/s41598-023-35177-6>
- Gelman, A., Vehtari, A., Simpson, D., Margossian, C. C., Carpenter, B., Yao, Y., Kennedy, L., Gabry, J., Bürkner, P.-C., & Modrák, M. (2020). *Bayesian Workflow* (No. arXiv:2011.01808). arXiv. <https://doi.org/10.48550/arXiv.2011.01808>
- Giersch, R. M., Hart, S. F. M., Reddy, S. G., Yonemitsu, M. A., Orellana Rosales, M. J., Korn, M., Geleta, B. M., Countway, P. D., Fernández Robledo, J. A., & Metzger, M. J. (2022). Survival and Detection of Bivalve Transmissible Neoplasia from the Soft-Shell Clam *Mya arenaria* (MarBTN) in Seawater. *Pathogens*, *11*(3), Article 3. <https://doi.org/10.3390/pathogens11030283>
- Grinsztajn, L., Semenova, E., Margossian, C. C., & Riou, J. (2021). Bayesian workflow for disease transmission modeling in Stan. *Statistics in Medicine*, *40*(27), 6209–6234. <https://doi.org/10.1002/sim.9164>
- Groner, M. L., Maynard, J., Breyta, R., Carnegie, R. B., Dobson, A., Friedman, C. S., Froelich, B., Garren, M., Gulland, F. M. D., Heron, S. F., Noble, R. T., Revie, C. W., Shields, J. D., Vanderstichel, R., Weil, E., Wyllie-Echeverria, S., & Harvell, C. D. (2016). Managing marine disease emergencies in an era of rapid change. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *371*(1689), 20150364. <https://doi.org/10.1098/rstb.2015.0364>
- Kermack, W. O., McKendrick, A. G., & Walker, G. T. (1997). A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character*, *115*(772), 700–721. <https://doi.org/10.1098/rspa.1927.0118>
- Lafferty, K. D. (2017). Marine Infectious Disease Ecology. *Annual Review of Ecology, Evolution, and Systematics*, *48*(Volume 48, 2017), 473–496. <https://doi.org/10.1146/annurev-eolsys-121415-032147>
- Leavitt, D. F., McDowell Capuzzo, J., Smolowitz, R. M., Miosky, D. L., Lancaster, B. A., & Reinisch, C. L. (1990). Hematopoietic neoplasia in *Mya arenaria*: Prevalence and indices of physiological condition. *Marine Biology*, *105*(2), 313–321. <https://doi.org/10.1007/BF01344301>

- McCallum, H. I., Kuris, A., Harvell, C. D., Lafferty, K. D., Smith, G. W., & Porter, J. (2004). Does terrestrial epidemiology apply to marine systems? *Trends in Ecology & Evolution*, 19(11), 585–591. <https://doi.org/10.1016/j.tree.2004.08.009>
- Metzger, M. J., & Goff, S. P. (2016). A Sixth Modality of Infectious Disease: Contagious Cancer from Devils to Clams and Beyond. *PLoS Pathogens*, 12(10), e1005904. <https://doi.org/10.1371/journal.ppat.1005904>
- Metzger, M. J., Reinisch, C., Sherry, J., & Goff, S. P. (2015). Horizontal Transmission of Clonal Cancer Cells Causes Leukemia in Soft-Shell Clams. *Cell*, 161(2), 255–263. <https://doi.org/10.1016/j.cell.2015.02.042>
- Murgia, C., Pritchard, J. K., Kim, S. Y., Fassati, A., & Weiss, R. A. (2006). Clonal origin and evolution of a transmissible cancer. *Cell*, 126(3), 477–487. <https://doi.org/10.1016/j.cell.2006.05.051>
- Pearse, A.-M., & Swift, K. (2006). Allograft theory: Transmission of devil facial-tumour disease. *Nature*, 439(7076), 549. <https://doi.org/10.1038/439549a>
- Tracy, A. M., Pielmeier, M. L., Yoshioka, R. M., Heron, S. F., & Harvell, C. D. (2019). Increases and decreases in marine disease reports in an era of global change. *Proceedings of the Royal Society B*. <https://doi.org/10.1098/rspb.2019.1718>
- Zwietering, M. H., Jongenburger, I., Rombouts, F. M., & van 't Riet, K. (1990). Modeling of the bacterial growth curve. *Applied and environmental microbiology*, 56(6), 1875–1881. <https://doi.org/10.1128/aem.56.6.1875-1881.1990>